

Docker or Virtual Machines: What's the Right Choice?

Whether you're a developer looking to streamline your deployment pipeline, a systems administrator managing infrastructure, or a technical decision-maker evaluating solutions for your organisation, this comparison of Docker and virtual machines aims to provide the insights needed to make the right choice for your use case.



In today's evolving technology landscape, developers and systems administrators face a common challenge: how to efficiently deploy, manage, and scale applications across different environments. Two primary solutions address this challenge: Docker containers and virtual machines (VMs). Both technologies serve the fundamental purpose of isolating applications and their dependencies, but they approach this goal through vastly different methods.

Understanding the distinction between Docker and VMs is crucial for making informed architectural decisions. While VMs have been the traditional approach for decades, containerisation

through Docker has revolutionised how we think about application deployment and resource utilisation.

What is Docker and Docker history

Docker is a containerisation platform that enables developers to package applications and their dependencies into lightweight, portable containers. These containers can run consistently across different environments, from development laptops to production servers, regardless of the underlying operating system or infrastructure.

At its core, Docker uses operating system-level virtualisation to create isolated user spaces called containers.

Unlike traditional virtualisation, containers share the host operating system's kernel while maintaining complete isolation of application processes, file systems, and network configurations. This approach makes containers significantly more resource-efficient than virtual machines.

Docker's journey began in 2008 when Solomon Hykes started working on a project called 'dotCloud'. The technology evolved from earlier containerisation concepts like Linux containers (LXC) but simplified the process dramatically. In March 2013, Docker was officially launched as an open source project, immediately gaining traction in the developer